

Algebra II with Trigonometry

Chapter 7.1

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4564659_alg-2trig-h-chp-7-1-note.pdf](#)

Here are 6 challenging, Olympiad-style mental math problems inspired by the identities, properties, and algebraic manipulations covered in the provided PDF. They are designed to be solved mentally (or with minimal scratch work) by rewarding insight, pattern recognition, and clever use of fundamental identities.

Problems

1. Evaluate the exact value of the following sum:
 $\log_2(\tan 1^\circ) + \log_2(\tan 2^\circ) + \log_2(\tan 3^\circ) + \cdots + \log_2(\tan 89^\circ)$
2. Suppose x is an angle in the interval $(0, \pi)$ such that $\csc x - \cot x = 5$. Determine the exact value of $\cos x$.
3. Find the exact numerical value of the following trigonometric expression:
 $\sin^6(2024^\circ) + \cos^6(2024^\circ) + 3 \sin^2(2024^\circ) \cos^2(2024^\circ)$
4. By considering an appropriate trigonometric substitution, determine the maximum possible value of the algebraic function $f(x) = \frac{x}{x^2+25}$ over all real numbers x .
5. Evaluate the exact value of the product:
 $(\sin^4(75^\circ) - \cos^4(75^\circ)) \sec(150^\circ)$
6. Given the equation $\frac{\sin x + \cos x}{\sin x - \cos x} = 3$, determine the exact value of $\sec^2 x$.